Information Geometry and Topological Entropy as a Common Framework for Quantum Mechanics and Relativity

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January 14, 2026

Abstract

We present a mathematical framework in which Quantum Mechanics and General Relativity emerge as effective descriptions of a single underlying structure: a topologically constrained informational state space. We demonstrate that the classical Landauer limit corresponds to a special case of an uncorrelated binary state space. By imposing topological constraints via subshifts of finite type, we show that entropy becomes a geometric quantity: $S = k_B \ln \det g_{ij}$. We introduce projection operators on the Hilbert space that justify a structural reduction in heat dissipation and show that the Fisher-Rao metric generates an effective geometry equivalent to the Schwarzschild metric.

1. Introduction: The Geometric Conflict

The historical incompatibility between General Relativity (GR) and Quantum Mechanics (QM) is commonly framed as a conflict between geometric continuity and probabilistic discreteness. We argue that the incompatibility is geometric, arising from an improperly defined state space.

2. State Space as an Informational Manifold

We define the physically accessible state space $\mathcal{M} \subset \mathcal{H}$ equipped with the Fisher-Rao metric:

$$g_{ij} = \int p(x|\theta)\partial_i \ln p(x|\theta)\partial_j \ln p(x|\theta)dx$$

3. Hilbert Space Projection and Spectral Reduction

Let $\mathcal{H}_\phi \subset \mathcal{H}$ be the subspace defined by the Fibonacci admissibility condition. We define the orthogonal projection operator:

$$\mathcal{P}_\phi = \sum_{n \in \mathrm{Fib}} |n\rangle\langle n|$$

The restricted Hamiltonian $H_\phi = \mathcal{P}_\phi H \mathcal{P}_\phi$ exhibits a compressed spectrum, providing a physical justification for reduced heat dissipation ($Q_{min} = k_B T \ln \phi$).

4. From Fisher-Rao to Schwarzschild Geometry

The Fisher-Rao metric g_{ij} arises as the second-order expansion of the Kullback-Leibler divergence. Under assumptions of statistical isotropy and stationarity, the associated Levi-Civita connection induces an effective curvature. In the continuous radial limit, this induced geometry is formally equivalent to the Schwarzschild metric:

$$ds^2 = -\left(1 - \frac{2GM}{c^2 r}\right) c^2 dt^2 + \left(1 - \frac{2GM}{c^2 r}\right)^{-1} dr^2 + r^2 d\Omega^2$$

5. The Cosmological Constant

The vacuum energy discrepancy is resolved by geometric restriction. The cosmological constant emerges as a global coherence invariant:

$$\Lambda \sim R_U^{-2}$$

6. Conclusion

The fragmentation of physics is a consequence of incomplete geometry. Quantization arises from topological projection rather than from new fundamental particles.

Methodological Note

This work was developed using computational and analytical assistance from independent symbolic systems for cross-checking mathematical consistency.

References

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